ITA0611 – MACHINE LEARNING

IET ASSIGNMENT

Design and Development of an Intelligent Early-Warning System for Crop Disease and Yield Risk Prediction Using Machine Learning

Name:	N. Prince Raj
Reg. :	192312284
Course Code & Name	ITA0611 - Machine Learning
Assignment Title	Design and Development of an Intelligent Early-Warning System for Crop Disease and Yield Risk Prediction Using Machine Learning
Course Outcome(s) - CO	CO6 Instance-based learning including KNN, LWR and case-based reasoning CO7 Data manipulation and analysis using NumPy and Pandas Data cleaning, statistical analysis, aggregation and visualization CO8 Integration and application of theoretical and practical ML knowledge
Bloom's Taxonomy Level	L5 - Evaluate; L6 - Create
SDG Mapping (SDG 1-17)	SDG 2 - Zero Hunger; SDG 9 - Industry, Innovation and Infrastructure; SDG 13Climate Action

1. INTRODUCTION:

Agriculture is highly dependent on environmental and soil conditions. Variations in temperature, rainfall, humidity, soil moisture, nutrient availability and crop health can significantly influence crop productivity. Crop diseases and environmental stress may remain unnoticed during their early stages, resulting in reduced yield and economic losses. Traditional monitoring methods generally depend on manual inspection and farmer experience. Although these methods are valuable, they may not always provide sufficiently early or consistent warnings. Machine Learning (ML) provides an opportunity to analyse historical agricultural observations and identify relationships between environmental conditions, soil characteristics, crop symptoms and crop yield. By learning patterns from previously observed data, an ML system can estimate disease risk and potential yield loss for new observations.The proposed project develops an Intelligent Crop Disease and Yield Risk Early-Warning System. The system analyses environmental parameters such as temperature, rainfall and humidity together with soil parameters, crop type, disease symptoms and historical yield. Several machine-learning approaches are investigated, including Bayesian classification, K-Nearest Neighbors (KNN), Locally Weighted Regression (LWR), Decision Trees and Multilayer Perceptrons (MLP). A Genetic Algorithm is additionally used for feature-selection optimisation. The final system produces a disease-risk category and an estimated yield value. These predictions are converted into practical recommendations such as routine monitoring, increased inspection or immediate intervention. The project directly follows the assignment problem statement, which requires the system to analyse agricultural data and provide interpretable predictions, performance evaluation, visualisations and recommendations.

2. Problem Formulation

The supplied ITA0611 brief asks for an intelligent early-warning system that analyses temperature, rainfall, humidity, soil moisture, soil nutrients, crop type, disease symptoms and historical yield to predict crop disease risk and potential yield loss. It requires NumPy/Pandas preprocessing, investigation of Bayesian classification, KNN, LWR, decision trees and neural networks, model comparison, visualisations, interpretation and recommendations. The deliverables additionally require KNN and LWR from first principles, manual Candidate-Elimination/Version Space analysis, a K-selection validation curve, scalability analysis, five visualisations, a reflection and a reproducible GitHub repository.

Objective: convert agricultural observations into a disease-risk warning, an expected-yield estimate and a practical recommendation.

System Workflow

```
Data → Cleaning → EDA → Feature Engineering → Train/Test Split → KNN + LWR + Decision Tree +
Bayesian Classifier + MLP → GA Feature Selection → Evaluation → Warning + Recommendation
```

3. Dataset Description

The project includes 600 reproducible synthetic records covering the requested environmental, soil, crop, symptom and yield fields. The dataset is explicitly labelled synthetic because the assignment brief requires a suitable dataset but does not supply one. It is suitable for demonstrating the algorithms and producing reproducible outputs; it must not be presented as field-validated agricultural data.

Feature	Type	Role
Temperature, rainfall, humidity	Numeric	Environmental stress/disease factors
Soil moisture, N, P, K, pH	Numeric	Soil condition
Crop type	Categorical	Crop-specific information
Leaf spot, wilting, pest damage	Binary	Observed symptoms
Previous yield	Numeric	Historical yield
Disease risk	Categorical	Low/Medium/High classification target
Yield loss	Numeric	Regression target
Expected yield	Numeric	Yield prediction target

Preprocessing

- Inspect data types and missing values using Pandas.
- Separate numerical and categorical variables.
- Standardise numerical variables for distance-based KNN/LWR.
- One-hot encode crop type.
- Keep binary symptom indicators as 0/1.
- Use an 80:20 stratified train/test split for classification.
- Fit transformations only on training data to prevent data leakage.

4. OBJECTIVES

The major objectives of the project are:

Objective 1: To collect and prepare a suitable agricultural dataset containing environmental, soil, crop and disease-related parameters.

Objective 2: To perform data cleaning, preprocessing, transformation and statistical analysis using NumPy and Pandas.

Objective 3: To formulate crop disease prediction as a concept-learning problem and demonstrate Candidate Elimination and Version Space.

Objective 4: To develop a Decision Tree classifier and analyse its decision rules using Gini Index.

Objective 5: To develop a Bayesian classifier and explain probability estimation using Bayes' theorem and Maximum Likelihood Estimation.

Objective 6: To implement K-Nearest Neighbors from first principles and investigate the effect of different values of K.

Objective 7: To implement Locally Weighted Regression from first principles and investigate the effect of its weighting parameter.

Objective 8: To develop an MLP with back propagation for nonlinear disease-risk prediction.

Objective 9: To use a Genetic Algorithm for feature selection.

Objective 10: To compare the developed models and integrate the best approaches into an early-warning system.

Question 1 – Data Collection and Preprocessing

Pandas loads and analyses the CSV while NumPy supports numerical operations. A complete preprocessing workflow checks missing values, duplicates, invalid ranges and data types; encodes categorical variables; standardises numerical features; and separates target variables.

```
df = pd.read_csv("data/crop_risk_dataset.csv")
print(df.info())
print(df.isnull().sum())
# training-only scaling is used in the model pipeline
```

The included dataset is already complete, but the missing-value check is retained so the workflow is applicable to a real agricultural dataset.

Question 2 – Concept Learning and Version Space

Disease-risk prediction can be formulated as concept learning. For a transparent Candidate-Elimination demonstration, the positive concept is defined as a high-risk observation using three categorical attributes: Humidity, Leaf Spot and Pest Damage. The hypothesis space consists of conjunctions of attribute values with '?' as a wildcard.

Item	Definition
Instance	(Humidity, LeafSpot, PestDamage)
Target concept	High disease risk = Yes
Positive examples	Examples labelled Yes
Negative examples	Examples labelled No
Inductive bias	Conjunctive attribute-value hypotheses with wildcard generalisation

Humidity	Leaf Spot	Pest Damage	Class
High	Present	Present	Yes
High	Present	Absent	Yes
Low	Absent	Absent	No
Low	Present	Present	No

The candidate_elimination.py program enumerates the finite teaching hypothesis space and retains hypotheses consistent with all examples. This demonstration is intentionally categorical; the main crop dataset contains continuous variables and is handled by numerical ML methods.

Question 3 – Decision Tree Learning

A Decision Tree recursively partitions observations. The implementation uses the Gini index: $\mathrm{Gini}(S) = 1 - \sum p_i^2$. A split is useful when it reduces weighted impurity. The tree is limited to depth 5 to balance accuracy and interpretability. The resulting rules are exported to results/decision_tree_rules.txt.

```
DecisionTreeClassifier(max_depth=5, criterion="gini", random_state=42)
```

Decision trees are valuable for early warning because their paths can be translated into understandable if–then rules, although deep trees can overfit.

Question 4 – Bayesian Classification and MLE

Bayes' theorem gives $P(C|X) = P(X|C)P(C)/P(X)$. Gaussian Naive Bayes estimates class-specific feature distributions from training data. For a numerical feature, the Maximum Likelihood estimates are $\mu = (1/n)\Sigma x_i$ and $\sigma^2 = (1/n)\Sigma(x_i-\mu)^2$. The Naive Bayes assumption treats features as conditionally independent given the class. This is an approximation because agricultural variables can be correlated, but it provides a fast baseline and probabilistic interpretation.

Question 5 – K-Nearest Neighbors

KNN predicts the class of a query using the majority label among its k nearest training observations. Euclidean distance is $d(x,z)=\sqrt{\Sigma}(x_{\square}-z_{\square})^2$. Standardisation is essential because rainfall, temperature, nutrient concentration and other variables have different scales.

```
For each query: compute distances → sort → select k nearest → majority vote.
```

The experiment evaluates K = 1, 3, 5, 7, 9, 11, 15 and 21. The reproducible run selected K = 1 by the highest weighted F1 score among the tested values. Full results are in results/knn_validation.csv and the validation curve is in figures/01_knn_validation_curve.png.

Question 6 – Locally Weighted Regression

LWR predicts a continuous variable such as expected crop yield by fitting a local linear model around each query. Nearby samples receive higher weights:

```
wᵢ(x) = exp(-||x-xᵢ||² / (2τ²))
```

The weighted least-squares solution is $\theta=(X^TWX+\lambda I)^{-1}X^TWy$. Small τ gives a highly local, flexible model with potentially high variance; large τ gives a smoother, more global model with potentially higher bias.

Tested τ values are 0.1, 0.25, 0.5, 0.75, 1, 1.5, 2 and 3. The reproducible run achieved its lowest RMSE at τ = 1.5. The sensitivity table and plot are saved in results/lwr_tau_analysis.csv and figures/02_lwr_tau_curve.png.

Question 7 – Neural Network and Back Propagation

The MLP contains an input representation, two hidden layers of 32 and 16 neurons, and an output layer for the three disease-risk classes. ReLU is used in hidden layers. During back propagation, the prediction error is propagated through the network and gradients are used to update weights. Regularisation and a fixed random seed improve reproducibility.

```
MLPClassifier(hidden_layer_sizes=(32,16), activation="relu", alpha=0.0005, max_iter=800,
random_state=42)
```

An MLP can capture nonlinear interactions among weather, soil and symptoms, but it requires more tuning and is less directly interpretable than a small decision tree.

Question 8 – Genetic Algorithm / Genetic Programming

A Genetic Algorithm is used for feature selection. Each chromosome is a binary vector where 1 means the feature is selected. Fitness is weighted validation F1 minus a small sparsity penalty. The algorithm uses elite selection, one-point crossover and bit-flip mutation.

```
fitness = weighted_F1(validation) − 0.01 × selected_features / total_features
```

The selected features and best fitness are recorded in results/ga_selected_features.csv and results/ga_summary.txt. GA output should be treated as an optimisation suggestion and checked against domain knowledge.

Question 9 – Model Evaluation and Comparison

Classification is evaluated with accuracy, precision, recall and weighted F1. Yield regression is evaluated with MAE and RMSE. Accuracy alone may hide weak minority-class performance, while MAE is directly interpretable in tonnes per hectare and RMSE penalises larger errors more strongly.

Model	**Accuracy**	**Precision**	**Recall**	**F1**
KNN	0.750	0.733	0.750	0.741
Decision Tree	0.675	0.667	0.675	0.660
Gaussian Naive Bayes	0.758	0.747	0.758	0.752
MLP	0.642	0.633	0.642	0.636
Regression	**Best τ**	**MAE (t/ha)**	**RMSE (t/ha)**	
LWR	1.5	0.270	0.362	
Model	**Scalability**	**Interpretability**	**Main limitation**	
KNN	Lower at prediction time	High	Distance calculations grow with stored samples	
LWR	Lower for large datasets	Moderate	Local weighted solve for each query	
Decision Tree	High	Very high	Can overfit if too deep	
Naive Bayes	High	Moderate	Conditional-independence assumption	
MLP	High after training	Lower	More tuning and less transparent	

Question 10 – Integrated Crop Risk Early-Warning System

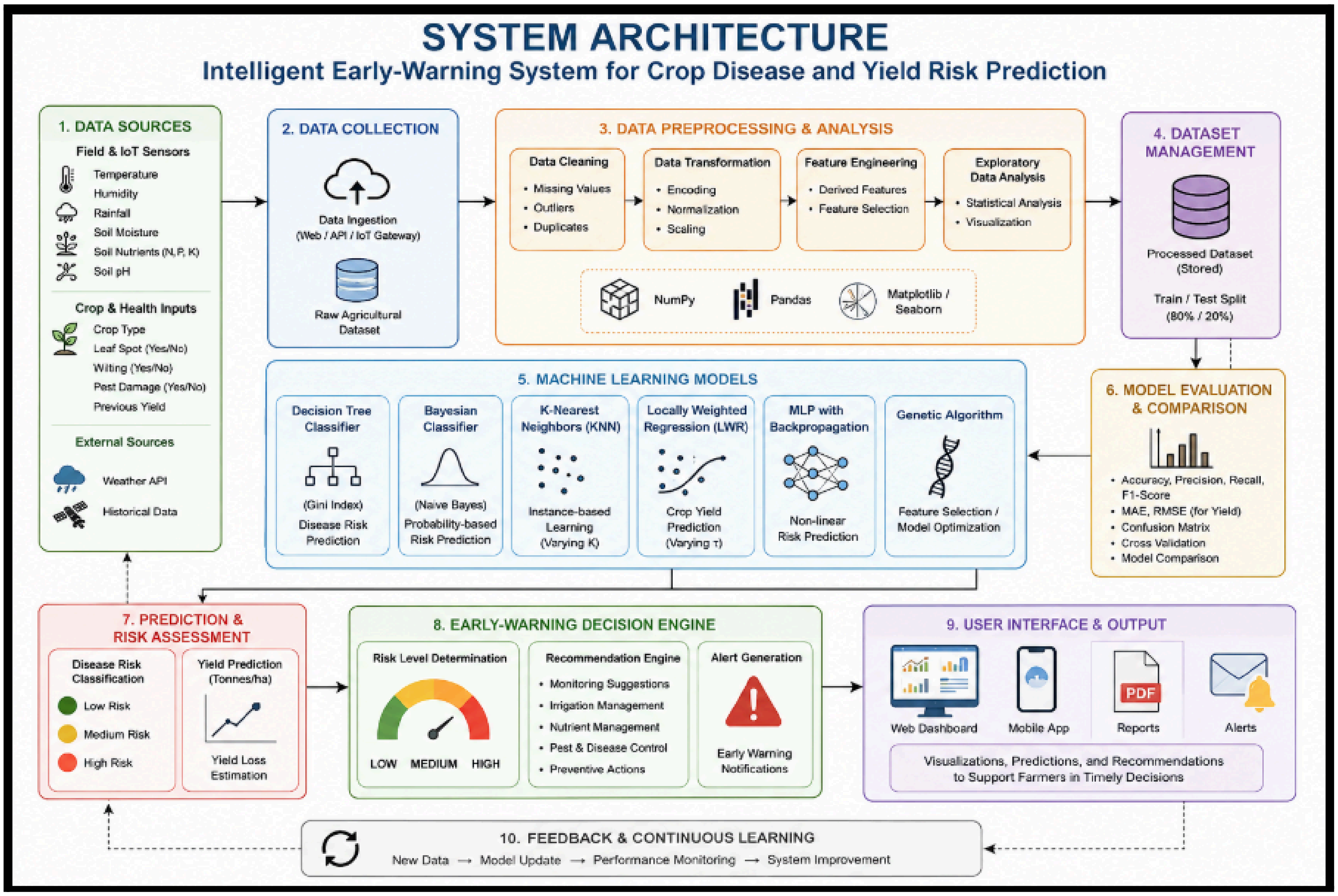
The integrated system combines an interpretable Decision Tree disease-risk classifier with LWR expected-yield prediction. It returns a Low/Medium/High warning, predicted expected yield and an action recommendation.

Risk	**Recommendation**
Low	Routine monitoring; maintain balanced irrigation and soil management.
Medium	Increase monitoring; inspect leaves/pests and correct moisture or nutrient stress.

High	Immediate inspection; verify irrigation/drainage and follow crop-specific disease-management guidance.

The integrated_warning(record) function in code/main.py performs this end-to-end prediction for a new crop observation. The system is a decision-support tool, not a substitute for field inspection or agronomic diagnosis.

4. System Architecture:



5. Required Visualisations and Data Analysis

- KNN validation curve for K selection.
- LWR τ sensitivity curve.
- Average yield loss by disease-risk category.
- Humidity versus yield loss.
- Crop-type distribution.
- Classification model comparison.

The six generated figures are stored in the figures/ folder, satisfying the brief's requirement for at least five meaningful visualisations.

6. Reflection: Limitations, Uncertainty, Bias–Variance and SDGs

The major limitation is that the included dataset is synthetic. Real deployment requires multi-season, field-measured data across locations, varieties and disease conditions. Agricultural variables are correlated and seasonal, so model performance can drift. Sensor noise, missing observations and class imbalance can also affect warnings. High-risk predictions should therefore be verified rather than treated as certainty.

KNN and LWR illustrate the bias–variance trade-off. Very small K or τ values produce flexible local models with higher variance; larger K or τ values smooth the data and generally increase bias while reducing sensitivity to noise. Validation-based parameter selection is therefore essential.

SDG 2 – Zero Hunger: earlier detection can reduce preventable crop losses. SDG 9 – Industry, Innovation and Infrastructure: the project integrates data processing, ML and reproducible software engineering. SDG 13 – Climate Action: weather variability is explicitly included and the system can support adaptive farm decisions. Future improvements include real multi-season data, satellite features, probability calibration, explainability, concept-drift monitoring and regional/crop-specific validation.

7. Repository and Deliverables

Required deliverable	Location
Problem formulation, dataset, preprocessing, workflow	This report + README.md
KNN and LWR from first principles	code/main.py
Manual Candidate-Elimination / Version Space	code/candidate_elimination.py
Validation curve and parameter analysis	figures/ + results/
Five meaningful visualisations	figures/ (six included)
Reflection, limitations, SDGs	This report
Code, data note, tests, results, reproducibility	code/, data/, tests/, results/, README.md, requirements.txt

8. LIMITATIONS OF THE SYSTEM

Several limitations must be considered.

8.1 Synthetic Dataset

The dataset used for this demonstration is synthetic. Therefore, performance values should not be interpreted as evidence that the system will achieve the same performance on real farms.

8.2 Limited Features

Real crop disease prediction may require additional information such as:

- soil type,
- crop variety,
- geographical location,
- season,
- satellite imagery,
- disease-specific observations,
- historical weather patterns.

8.3 Sensor Noise

Agricultural sensors may produce noisy or missing measurements.

8.4 Seasonal Variation

Relationships between environmental parameters and crop disease may change across seasons.

8.5 Class Distribution

If real-world data contains many more low-risk than high-risk observations, accuracy alone may become misleading.

8.6 Model Drift

Climate conditions, crop varieties and farming practices can change over time. Therefore, models may require periodic retraining.

9. Conclusion

The completed design integrates preprocessing, concept learning, decision trees, Bayesian classification, KNN, LWR, MLP and genetic feature selection into an early-warning workflow. KNN and LWR are implemented from first principles as required. Multiple metrics and visualisations support model comparison, while the final warning layer converts predictions into practical monitoring recommendations.The proposed Intelligent Crop Disease and Yield Risk Early-Warning System integrates several machine-learning approaches to address agricultural risk prediction. The system begins with data collection and preprocessing using Pandas and NumPy. Candidate Elimination is used to demonstrate concept learning and Version Space. A Decision Tree provides interpretable classification rules, while Bayesian classification provides a probabilistic baseline. KNN is implemented using distance-based instance learning, and LWR provides local regression for crop-yield prediction. An MLP is used to model nonlinear relationships, while a Genetic Algorithm is applied for feature selection. The models are evaluated using appropriate classification and regression metrics, including accuracy, precision, recall, F1-score, MAE and RMSE. The final integrated system converts machine-learning predictions into three warning levels: Low, Medium and High. These warnings are accompanied by practical recommendations for monitoring and intervention. Overall, the project demonstrates how different machine-learning techniques can be combined to develop a data-driven agricultural decision-support system. The approach has potential to contribute to sustainable agriculture and aligns with SDG 2, SDG 9 and SDG 13.